Reviewer 4 Reproducibility Supplement

State of Health Estimation and Target Leakage Auditing for Lithium Ion Batteries

The manuscript retains all prior scientific evidence and adds conditional diagnostics. CSVs are authoritative numerical records; this supplement provides readable definitions, complete NASA battery memberships, stopping summaries and regeneration instructions. Earlier model artifacts are not overwritten.

S1 Feature and preprocessing definitions

Proxy definitions: for retained rows k sorted within each battery, retention = Q_k/Q_ref; cumulative = sum of Q_1 through Q_k; difference = Q_k−Q_(k−1), zero at the first row; rolling SOH = arithmetic mean of the current and previous four retained SOH rows, with minimum one. All are current-target proxies and audit-only. The 18 observable features do not include any of these columns.

Observable scalar order: cycle_index; prior_equivalent_full_cycles; ambient_temperature; segment_duration_s; segment_energy_wh; voltage_start_v; voltage_end_v; voltage_mean_v; voltage_std_v; voltage_linear_slope_v_per_s; voltage_curvature; current_mean_c_rate; current_std_c_rate; temperature_mean_c; temperature_std_c; temperature_rise_c; temperature_slope_c_per_s; temperature_observed.

Prior EFC accumulates capacities strictly preceding the current cycle and divides by published rated Ah (NASA 2.0, CALCE 1.1, Oxford 0.74). Duration is final minus initial segment time. Partial energy = sum(|I_i| V_i max(Δt_i,0))/3600 Wh. Voltage/temperature/current means and standard deviations are computed on the selected segment, with ddof=0. Current loading uses absolute current/rated Ah. Linear slopes are least-squares slopes versus elapsed seconds. Curvature is the quadratic voltage coefficient versus normalized elapsed time. Thermal rise is max(T)−min(T). Observation status indicates whether a thermal channel was present.

Sequence channels are normalized time, voltage, absolute C-rate, temperature and thermal observation status, sampled at 64 points on [0,1] by linear interpolation. The segment begins at the first measured voltage ≤4.0 V and ends at the first subsequent voltage ≤3.6 V. At least three samples are required, and sequence duration must be positive. No temperature/rate-specific boundary is fitted.

Preprocessing: source-trained scalar median imputation; StandardScaler for SVR/Ridge/KNN only; sequence median imputation followed by source-trained channel mean and standard deviation; neural target mean/std from training rows. Missing sequence values are linearly interpolated within the completed segment with endpoint filling before resampling. Entirely absent thermal traces use ambient fallback with indicator 0. These fixed per-record operations differ from fitting a normalization statistic on external data.

Source: application/src/data/partial_cycle.py; application/src/experiments/classical.py; application/src/experiments/deep.py; application/scripts/run_protocol_comparison.py. Current application APIs and model defaults are outside this manuscript revision.

S2 Complete NASA split membership and support

Exactly the saved split identities are reproduced for each seed. Every fold has 19 train, five validation and six test cells. Class order in the tables is Healthy / Aging / Near-EOL / EOL. Quartiles and all row-level records are in the accompanying CSVs. No outcome stratification or post-score balancing is claimed.

Seed 17

Supplement S2. Seed 17 complete role inventory

Fold / role	Battery IDs	Cycles	SOH min / median / max	H / A / N / E cycles
1 train	B0007; B0018; B0025; B0027; B0029; B0030; B0031; B0033; B0038; B0039; B0040; B0042; B0043; B0045; B0047; B0048; B0053; B0055; B0056	1196	65.4 / 89.9 / 116.6	594 / 366 / 216 / 20
1 validation	B0006; B0028; B0034; B0036; B0054	522	57.0 / 91.4 / 110.7	293 / 102 / 64 / 63
1 test	B0005; B0026; B0032; B0041; B0044; B0046	423	69.7 / 89.0 / 114.0	202 / 105 / 114 / 2
2 train	B0006; B0025; B0026; B0027; B0029; B0030; B0031; B0032; B0036; B0038; B0039; B0040; B0041; B0043; B0044; B0045; B0046; B0055; B0056	1234	57.0 / 91.6 / 116.6	703 / 287 / 161 / 83
2 validation	B0005; B0028; B0033; B0034; B0048	317	69.7 / 91.0 / 113.8	162 / 82 / 71 / 2
2 test	B0007; B0018; B0042; B0047; B0053; B0054	590	72.6 / 86.7 / 109.8	224 / 204 / 162 / 0

Fold / role	Battery IDs	Cycles	SOH min / median / max	H / A / N / E cycles
3 train	B0007; B0018; B0025; B0026; B0028; B0032; B0033; B0034; B0040; B0041; B0042; B0043; B0044; B0045; B0046; B0047; B0048; B0054; B0056	1264	65.4 / 89.0 / 116.6	595 / 387 / 262 / 20
3 validation	B0005; B0027; B0038; B0039; B0053	293	69.7 / 91.8 / 100.6	166 / 61 / 64 / 2
3 test	B0006; B0029; B0030; B0031; B0036; B0055	584	57.0 / 91.4 / 110.0	328 / 125 / 68 / 63
4 train	B0006; B0007; B0018; B0026; B0029; B0030; B0031; B0032; B0039; B0040; B0041; B0042; B0044; B0045; B0046; B0047; B0053; B0055; B0056	1372	57.0 / 88.2 / 116.6	603 / 389 / 297 / 83
4 validation	B0005; B0027; B0036; B0038; B0054	499	69.7 / 92.0 / 110.0	299 / 117 / 81 / 2
4 test	B0025; B0028; B0033; B0034; B0043; B0048	270	75.3 / 96.6 / 113.8	187 / 67 / 16 / 0
5 train	B0006; B0007; B0018; B0025; B0028; B0029; B0030; B0031; B0034; B0036; B0041; B0042; B0043; B0044; B0046; B0047; B0048; B0054; B0055	1526	57.0 / 89.5 / 114.0	734 / 433 / 296 / 63
5 validation	B0005; B0026; B0032; B0033; B0053	341	69.7 / 92.7 / 113.8	207 / 68 / 64 / 2
5 test	B0027; B0038; B0039; B0040; B0045; B0056	274	65.4 / 91.0 / 116.6	148 / 72 / 34 / 20

Seed 42

Supplement S2. Seed 42 complete role inventory

Fold / role	Battery IDs	Cycles	SOH min / median / max	H / A / N / E cycles
1 train	B0006; B0007; B0018; B0025; B0026; B0027; B0028; B0032; B0033; B0036; B0039; B0041; B0042; B0044; B0047; B0048; B0053; B0055; B0056	1480	57.0 / 90.4 / 113.8	761 / 409 / 247 / 63
1 validation	B0005; B0031; B0034; B0043; B0045	345	65.4 / 86.7 / 116.6	144 / 78 / 101 / 22
1 test	B0029; B0030; B0038; B0040; B0046; B0054	316	74.1 / 91.9 / 114.0	184 / 86 / 46 / 0
2 train	B0007; B0018; B0026; B0027; B0028; B0029; B0030; B0032; B0034; B0038; B0040; B0041; B0042; B0044; B0046; B0048; B0053; B0054; B0056	1152	72.8 / 90.5 / 114.0	604 / 362 / 186 / 0
2 validation	B0006; B0031; B0036; B0043; B0045	538	57.0 / 90.8 / 116.6	284 / 87 / 84 / 83
2 test	B0005; B0025; B0033; B0039; B0047; B0055	451	69.7 / 86.9 / 113.8	201 / 124 / 124 / 2
3 train	B0018; B0025; B0027; B0028; B0029; B0030; B0031; B0036; B0038; B0040; B0041; B0042; B0043; B0046; B0048; B0053; B0054; B0055; B0056	1278	72.8 / 91.6 / 114.0	734 / 383 / 161 / 0
3 validation	B0005; B0033; B0039; B0044; B0047	359	69.7 / 87.9 / 113.8	160 / 87 / 110 / 2
3 test	B0006; B0007; B0026; B0032; B0034; B0045	504	57.0 / 84.6 / 116.6	195 / 103 / 123 / 83
4 train	B0006; B0007; B0025; B0026; B0027; B0028; B0029; B0031; B0032; B0034; B0036; B0038; B0039; B0042; B0045; B0046; B0047; B0054; B0055	1366	57.0 / 91.3 / 116.6	736 / 310 / 237 / 83
4 validation	B0005; B0030; B0033; B0040; B0043	328	69.7 / 92.1 / 113.8	186 / 71 / 69 / 2
4 test	B0018; B0041; B0044; B0048; B0053; B0056	447	72.8 / 87.8 / 110.0	167 / 192 / 88 / 0
5 train	B0006; B0007; B0018; B0025; B0026; B0029; B0030; B0033; B0034; B0039; B0040; B0041; B0044; B0046; B0048; B0053; B0054; B0055; B0056	1338	57.0 / 88.5 / 114.0	598 / 427 / 250 / 63
5 validation	B0005; B0032; B0038; B0045; B0047	380	65.4 / 84.2 / 116.6	149 / 78 / 131 / 22
5 test	B0027; B0028; B0031; B0036; B0042; B0043	423	75.3 / 94.7 / 110.0	342 / 68 / 13 / 0

Seed 2026

Supplement S2. Seed 2026 complete role inventory

Fold / role	Battery IDs	Cycles	SOH min / median / max	H / A / N / E cycles
1 train	B0005; B0006; B0007; B0025; B0026; B0027; B0028; B0030; B0032; B0033; B0034; B0038; B0039; B0043; B0048; B0053; B0054; B0055; B0056	1338	57.0 / 89.8 / 113.8	658 / 402 / 213 / 65
1 validation	B0018; B0040; B0041; B0042; B0047	324	72.6 / 86.5 / 109.8	133 / 78 / 113 / 0
1 test	B0029; B0031; B0036; B0044; B0045; B0046	479	65.4 / 92.3 / 116.6	298 / 93 / 68 / 20
2 train	B0005; B0006; B0007; B0026; B0027; B0028; B0029; B0031; B0033; B0034; B0036; B0039; B0040; B0045; B0047; B0048; B0053; B0054; B0056	1450	57.0 / 90.6 / 116.6	756 / 357 / 252 / 85

Fold / role	Battery IDs	Cycles	SOH min / median / max	H / A / N / E cycles
2 validation	B0025; B0042; B0043; B0044; B0046	320	74.1 / 91.6 / 114.0	175 / 96 / 49 / 0
2 test	B0018; B0030; B0032; B0038; B0041; B0055	371	72.8 / 87.7 / 105.9	158 / 120 / 93 / 0
3 train	B0006; B0007; B0018; B0027; B0029; B0030; B0031; B0032; B0033; B0034; B0036; B0038; B0041; B0045; B0047; B0048; B0053; B0054; B0055	1372	57.0 / 88.8 / 116.6	632 / 378 / 279 / 83
3 validation	B0026; B0042; B0043; B0044; B0046	320	74.1 / 91.4 / 114.0	173 / 96 / 51 / 0
3 test	B0005; B0025; B0028; B0039; B0040; B0056	449	69.7 / 95.4 / 102.0	284 / 99 / 64 / 2
4 train	B0005; B0006; B0018; B0027; B0028; B0029; B0030; B0031; B0032; B0034; B0036; B0038; B0039; B0045; B0047; B0048; B0053; B0055; B0056	1416	57.0 / 89.9 / 116.6	705 / 360 / 266 / 85
4 validation	B0025; B0040; B0041; B0044; B0046	248	72.9 / 93.0 / 114.0	140 / 58 / 50 / 0
4 test	B0007; B0026; B0033; B0042; B0043; B0054	477	74.5 / 90.4 / 113.8	244 / 155 / 78 / 0
5 train	B0005; B0007; B0018; B0026; B0028; B0029; B0030; B0031; B0032; B0033; B0036; B0038; B0039; B0043; B0045; B0046; B0054; B0055; B0056	1532	65.4 / 91.0 / 116.6	823 / 413 / 274 / 22
5 validation	B0025; B0040; B0041; B0042; B0044	244	72.9 / 95.5 / 105.9	161 / 54 / 29 / 0
5 test	B0006; B0027; B0034; B0047; B0048; B0053	365	57.0 / 82.7 / 110.7	105 / 106 / 91 / 63

S3 Matched control design and results

Each design uses 21 split/seed instances and three contracts (63 fits). Test rows are unchanged within each paired comparison. Equal-count design uses 1,152 training rows. Exposure-matched sensitivity uses common temperature quotas 416 at 4 °C, 380 at 24 °C and 40 at 43 °C; zero quotas at 22/44 °C do not remove held-out tests. Fixed ExtraTrees parameters: 300 trees, minimum leaf 2, random state = seed. Shuffled columns draw donors independently inside train or test partitions. They preserve column marginals and count without transmitting held-out values to training.

Equal-count results

Protocol	Contract	Fold mean MAE	Pooled sample MAE	Battery macro MAE
v1 random cycle	18 observable	0.860	0.844	1.130
v1 random cycle	18 + proxies	0.093	0.087	0.106
v1 random cycle	18 + sham	0.889	0.874	1.195
v2 chronological	18 observable	2.652	2.633	3.786
v2 chronological	18 + proxies	0.473	0.463	0.419
v2 chronological	18 + sham	2.690	2.671	3.865
v3 grouped battery	18 observable	4.171	4.165	3.965
v3 grouped battery	18 + proxies	0.439	0.435	0.404
v3 grouped battery	18 + sham	4.181	4.167	3.997

Common-temperature-quota results

Protocol	Contract	Fold mean MAE	Pooled sample MAE	Battery macro MAE
v1 random cycle	18 observable	1.149	1.114	1.602
v1 random cycle	18 + proxies	0.102	0.096	0.143
v1 random cycle	18 + sham	1.167	1.134	1.619
v2 chronological	18 observable	2.640	2.615	3.630
v2 chronological	18 + proxies	0.477	0.469	0.370
v2 chronological	18 + sham	2.708	2.680	3.642
v3 grouped battery	18 observable	4.190	4.191	3.980
v3 grouped battery	18 + proxies	0.429	0.418	0.416
v3 grouped battery	18 + sham	4.189	4.175	3.961

Stratified error CSVs retain every observed ambient temperature and fixed C-rate bands <0.75, 0.75–1.25 and ≥1.25 C. Counts and distinct-cell support accompany MAE. An absent stratum is not a zero error. V1 tests cover only sampled

cycles; V2 tests cover late retained cycles; V3 tests cover all cells across grouped folds. Cross-protocol pooled metrics therefore do not share one test population.

S4 Deep stopping and validation diagnostics

The original release supplies 420 grouped-fit epoch totals and training/test scores: NASA 150, CALCE 120, Oxford 150. Each model's executed epochs include patience. Original best-epoch identifiers and validation histories were not retained and cannot be reconstructed honestly. The secondary NASA reruns supply separate validation/test predictions for 150 fits, with full per-fit gaps in secondary_nasa_validation_test_gaps.csv. These do not replace original regression results. Ordinary PyTorch models monitor normalized mean-batch MAE; VAE-LSTM monitors its combined objective; TensorFlow physics-iTransformer monitors weighted two-head loss. Early stopping restores the best monitored-loss state, with a 200-epoch cap and patience 20.

Original NASA stopping summary

Model	Fits	Epoch min / med / max	Train MAE	Test MAE	Test − train
Attention LSTM	15	21/45/94	4.922	8.171	3.249
BatteryGPT	15	22/39/99	4.275	8.475	4.200
Bidirectional LSTM	15	21/52/89	4.468	6.930	2.463
Dynamic-Graph iTransformer	15	21/44/91	3.161	6.800	3.639
GRU	15	21/32/85	5.963	7.619	1.656
iTransformer	15	21/39/56	3.231	7.295	4.065
Physics-Informed iTransformer	15	21/33/78	3.497	7.533	4.036
Temporal Fusion Transformer	15	21/53/120	4.547	7.658	3.110
VAE-LSTM	15	25/56/116	4.134	7.790	3.656
Vanilla LSTM	15	22/28/52	5.854	7.494	1.640

Original Calce stopping summary

Model	Fits	Epoch min / med / max	Train MAE	Test MAE	Test − train
Attention LSTM	12	39/78/107	2.967	4.385	1.418
BatteryGPT	12	38/60.5/108	3.518	5.713	2.195
Bidirectional LSTM	12	31/76.5/105	3.529	6.479	2.950
Dynamic-Graph iTransformer	12	25/48.5/63	3.046	5.239	2.193
GRU	12	50/97.5/162	2.735	3.739	1.005
iTransformer	12	34/49.5/67	2.613	5.249	2.636
Physics-Informed iTransformer	12	26/56.5/85	2.672	5.539	2.866
Temporal Fusion Transformer	12	59/75/112	3.439	4.993	1.554
VAE-LSTM	12	22/72.5/114	5.235	7.600	2.365
Vanilla LSTM	12	22/79/200	6.542	7.094	0.552

Original Oxford stopping summary

Model	Fits	Epoch min / med / max	Train MAE	Test MAE	Test − train
Attention LSTM	15	21/30/85	4.778	5.778	1.001
BatteryGPT	15	21/39/81	2.707	4.681	1.974
Bidirectional LSTM	15	22/43/199	3.136	4.167	1.031
Dynamic-Graph iTransformer	15	21/60/149	2.079	3.970	1.892
GRU	15	21/56/108	2.514	3.669	1.154
iTransformer	15	26/56/157	1.179	2.209	1.031
Physics-Informed iTransformer	15	25/84/144	1.033	2.026	0.993
Temporal Fusion Transformer	15	21/38/123	3.314	5.600	2.286
VAE-LSTM	15	28/44/119	3.154	5.588	2.434
Vanilla LSTM	15	21/45/200	3.117	4.541	1.424

Secondary NASA validation/test diagnostics

Model	Epoch min / med / max	Validation MAE	Test MAE	Test − validation
Attention LSTM	21/38/96	5.518	7.472	1.954
BatteryGPT	21/40/94	4.966	8.496	3.529
Bidirectional LSTM	21/48/111	4.943	7.550	2.607
Dynamic-Graph iTransformer	22/35/103	4.215	7.076	2.861
GRU	21/32/96	6.338	7.538	1.200
iTransformer	21/27/96	4.535	6.473	1.938
Physics-Informed iTransformer	21/27/96	4.535	6.473	1.938
Temporal Fusion Transformer	21/29/112	5.438	8.406	2.968
VAE-LSTM	25/39/111	5.240	8.317	3.077
Vanilla LSTM	22/34/59	6.269	7.777	1.507

S5 Metric aggregation and ensemble fit isolation

Average the three held-out predictions per cycle before computing headline errors. Sample MAE weights cycles equally; battery-macro MAE averages each cell's MAE equally; fold-mean MAE averages fit-level MAEs. The CSV nasa_metric_estimands.csv gives all three for all 20 models. Ensemble fit records identify disjoint roles; weights are computed per seed-fold using validation records, then frozen for testing. Full-source frozen transfer has its own NASA-only out-of-fold meta fit. No external label enters fitting.

NASA estimands on preserved regression artifacts

Model	Battery-macro MAE	Pooled sample MAE	Mean fold MAE
Attention LSTM	6.668	7.936	8.171
BatteryGPT	6.714	8.398	8.475
Bidirectional LSTM	5.946	6.744	6.930
Dynamic-Graph iTransformer	5.097	6.387	6.800
Extra Trees	3.956	4.146	4.145
Gradient Boosting	4.054	3.957	4.157
GRU	6.260	7.596	7.619
iTransformer	5.465	6.873	7.295
K-Nearest Neighbours	4.036	4.773	4.825
LightGBM	3.735	3.865	4.072
Physics-Informed iTransformer	5.783	6.991	7.533
Random Forest	3.832	3.827	3.963
Ridge Regression	6.975	5.840	6.529
Stacking Ensemble	3.802	4.002	4.647
Support Vector Regression	3.628	3.752	3.931
Temporal Fusion Transformer	6.045	7.354	7.658
VAE-LSTM	6.583	7.486	7.790
Vanilla LSTM	6.161	7.547	7.494
Weighted Average Ensemble	3.536	3.682	3.779
XGBoost	3.892	3.962	4.079

S6 Inference and small-cell sensitivity

NASA signed-rank inference uses 30 paired cell scores after repeat averaging. Holm adjusts one family of 19 two-sided reference-versus-other comparisons with zero_method=zsplit. Selection of the reference from the same benchmark is not corrected, so significance remains exploratory. Cluster intervals condition on predictions and observed cells, omit model-retraining uncertainty, and do not guarantee unconditional population coverage. Four CALCE cells cannot support robust population-level model superiority; folds are not independent extra cells.

CALCE cell influence without model refitting

Omitted battery	Remaining cells	Cycles	Sample MAE	Battery macro MAE
CS2_35	3	2911	1.222	1.234
CS2_36	3	2880	0.968	0.961
CS2_37	3	2803	1.345	1.339
CS2_38	3	2773	1.067	1.079

S7 Permutation and retraining sensitivity

Five seed-42 grouped fits supply 2,141 held-out cycles from 30 cells. Each permutation feature/block uses ten fixed donor permutations; group columns share donors. Bootstrap 10,000 whole-cell draws, weighting the repeated cells' errors and counts, produces conditional pointwise intervals. Pooled donors mix test cells; within-battery donors preserve exposure identity. The interaction contrast $\Delta(T,V)-\Delta(T)-\Delta(V)$ uses the same donor construction and is not a causal or multiplicity-adjusted finding. Individual-feature, group and interaction records are saved. Temperature ablation retrains without all six thermal columns and compares the same test cycles. Its pooled effect −0.130 pp, interval −0.352 to 0.133, is inconclusive.

S8 Suggested-reference assessment

Haotian Zhou. Research on the Fusion Model of DeepFM and XGBoost for Digital Consumer Behavior Prediction. 10.71451/ISTAER2617. discussed in response, not added to manuscript. Consumer click prediction concerns a different endpoint and validation unit; it does not substantiate battery SOH validity or charging optimization. https://istaer.online/index.php/Home/article/view/No.2617

Xi Zeng. Cross-Border Trade Fraud Detection via Integrated Heterogeneous Graph Neural Network and XGBoost. 10.71451/ISTAER2603. discussed in response, not added to manuscript. Fraud classification does not address battery observability, held-out cells or charging control. https://istaer.online/index.php/Home/article/view/No.2603

Guona Chen. Research on an Energy Economic Demand Time Series Prediction Model Based on the Integration of XGBoost and Deep Learning. 10.67541/jdc2601. added as limited methodological context, reference 26. Energy-demand forecasting uses tree/deep fusion; the manuscript explicitly distinguishes this forecast endpoint from retrospective battery SOH estimation. No numerical results transferred. https://dipscie.com/index.php/JDC/article/view/10.67541_jdc2601

Wenhao Wang, Siyi Shen, Ye Yuan. Forecasting Short-Term Export Volumes with Hybrid Models Integrating SARIMA with Attention-Based LSTM. 10.71451/ISTAER2601. discussed in response, not added to manuscript. Export forecasting is outside the battery estimation endpoint. Publisher lists Yuan Y as third author, whereas the review lists Wang Y. https://istaer.online/index.php/Home/article/view/No.2601

Guona Chen. Research on Algorithm Improvement of ARIMA-LSTM Hybrid Model in Time Series Prediction of Inflation Rate. 10.71451/ISTAER2605. discussed in response, not added to manuscript. Inflation time-series forecasting does not provide battery-specific estimation or validation evidence. https://istaer.online/index.php/Home/article/view/No.2605

S9 Regeneration and provenance

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application/.venv/Scripts/python.exe scripts/paper/r4_test_analysis.py

application/.venv/Scripts/python.exe scripts/paper/r4_analysis.py all

application/.venv/Scripts/python.exe scripts/paper/r4_figures.py

application/.venv/Scripts/python.exe scripts/paper/r4_verify_science.py

BUNDLED_PYTHON scripts/paper/r4_build_documents.py

BUNDLED_PYTHON scripts/paper/r4_verify_documents.py
```

Run from the workspace root. BUNDLED_PYTHON denotes the document runtime resolved through the Codex workspace dependency loader. The scientific runtime is the existing application virtual environment. Output paths stay in paper/revision-2026-11-r4; archived regression/calibration artifacts are read-only. Figure files are vector PDF/SVG plus 600-dpi PNG. Source hashes, test output and verification reports document the exact inputs used. No deployment or publication is performed by these commands.